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POLICY BRIEF

The Developmental State in the Quantum Era: Japan's Quantum Computing Strategy Between Industrial Policy and National Security

Summary

Quantum computing is emerging as a transformative technology with potential to reshape the global balance of economic power and the foundations of national security in the twenty-first century. By exploiting the principles of quantum mechanics (i.e. superposition, entanglement and interference), quantum computers promise exponential gains in processing power relative to classical computing. These advances have far-reaching implications for encryption, secure communication, advanced materials, pharmaceuticals, logistics and artificial intelligence.

For Japan, the stakes are especially high. The country has a strong legacy of industrial policy and technological achievement – semiconductors in the 1970s, consumer electronics in the 1980s and then robotics – but its position has weakened facing U.S. dominance in cutting edge technology, particularly artificial intelligence, and the state-driven technological ascent of China, including its control over rare earth minerals, which are essential in semiconductor supply chains. Against this backdrop, quantum computing is not simply another high-tech sector but a field in which Japan's broader economic security strategy is being tested.

This brief argues that Japan's quantum strategy is an example of the persistence and adaptation of the developmental state model in an

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era of securitised technology policy. Just as in the 1970s the Ministry of International Trade and Industry (MITI) organised consortia to close the gap in semiconductors, today policymakers in the Cabinet Office, the Ministry of Defence (MOD), the Ministry of Economy, Trade and Industry (METI) and the Ministry of Education, Culture, Sports, Science and Technology (MEXT) are structuring public-private collaboration in quantum computing. Unlike earlier projects that focused primarily on industrial competitiveness and growth, the contemporary model explicitly links technological development with national security, which reflects the pressure of U.S.-China rivalry and the country's heightened economic security agenda. Ultimately, Japan's quantum strategy is an example of a state with limited resources but a strong developmental tradition seeking to sustain strategic relevance in an era characterised by competition over critical technologies.

1. Introduction[1]

Quantum computing has drawn growing attention not only by academia but also by policymakers, as it has become a central area of geopolitical competition. As Chou, Manyika and Neven recently observed in *Foreign Affairs*, the race to lead in quantum technologies is no longer only about scientific prestige but it has become a contest over economic security, military advantage and technological sovereignty.[2] For the United States and China, this race is a key part of their broader rivalry: Washington designates quantum as a "critical and emerging technology"[3] essential for national defence, while Beijing has invested billions in quantum communication networks and satellite experiments as part of its techno-nationalist strategy. For Japan, this global contest involves both urgency and vulnerability. The designation of 2025 by the Ishiba administration as the "first year of quantum industrialisation" (*ryōshi sangyōka gannen*) signals Tokyo's determination to avoid marginalisation in a field increasingly tied to national security and alliance politics.[4]

The urgency of quantum computing stems most directly from its national security implications. Quantum algorithms such as Shor's could make widely used encryption protocols (such as RSA and ECC) obsolete, and create vulnerabilities in defence communication, financial systems and digital infrastructure. Conversely, quantum key distribution (QKD) promises unbreakable encryption for diplomatic and military communication, while quantum sensors may detect stealth platforms and submarines. Quantum-enhanced simulations also have potential for breakthroughs in material science and weapon development. For Japan, these possibilities directly intersect with its strategic vulnerabilities: proximity to nuclear-armed North Korea, Russia and China; reliance on maritime trade routes; and dependence on U.S. extended deterrence.

Japan's pursuit of quantum technologies must also be understood in a historical perspective. The developmental state tradition pioneered by Chalmers Johnson in *MITI and the Japanese Miracle* emphasised the role of the state in coordinating industrial policy, nurturing state-industry linkages and steering long-term technological growth with powerful pilot agencies.[5] Subsequent scholarship extended this framework to fields such as high technology and biotechnology,[6] but it has largely remained focused on economic catch-up and industrial development rather than national security.

1 This research was supported by JSPS Grants-in-Aid for Scientific Research (25K04956).

2 Charina Chou, James Manyika and Hartmut Neven, "The Race to Lead the Quantum Future," *Foreign Affairs* 104, no. 1 (2025): 154-67

3 US National Science and Technologies Council, Critical and Emerging Technologies List Update, February 2024, https://www.govinfo.gov/content/pkg/CMR-PREX23-00185928/pdf/CMR-PREX23-00185928.pdf/.

4 *Asahi Shimbun*, 3 June 2025, https://digital.asahi.com/articles/DA3S16227410.html.

5 Chalmers Johnson, *MITI and the Japanese Miracle: The Growth of Industrial Policy, 1925-1975* (Stanford, CA: Stanford University Press, 1982).

6 See Daniel I. Okimoto, *Between MITI and the Market: Japanese Industrial Policy for High Technology* (Stanford, CA: Stanford University Press, 1989); Marie Anchordoguy, *Computers, Inc.: Japan's Challenge to IBM* (Cambridge, MA: Harvard University Press, 1989); and Joseph Wong, *Betting on Biotech: Innovation and the Limits of Asia's Developmental State* (Ithaca, NY: Cornell University Press, 2011).

A notable exception is Linda Weiss's *America Inc?*,[7] which demonstrates how the U.S. security state has steered R&D in critical technologies. Japan's emerging quantum strategy suggests a similar extension of developmental state practices into the security domain. This dynamic is not entirely new. The Abe government's relaxation of arms export restrictions in 2014 and its promotion of defence industrial cooperation likewise signalled stronger state leadership in fostering innovation in military technology with enhanced government support.[8] Today, policymakers working in the Cabinet Office, MOD, METI and MEXT are orchestrating public-private collaboration in quantum computing, not only to stimulate innovation but also to strengthen resilience and strategic autonomy. Thus, Japan's quantum policy represents both a renewal of developmental state practices and an adaptation of that model into a securitised developmental state, in which technological policy is framed simultaneously in terms of industrial competitiveness and national survival.

Building on this perspective, this brief argues that Japan's quantum strategy exemplifies the persistence of a developmental state, now reconfigured into a securitised developmental state that fuses industrial policy with national security. The analysis proceeds in four steps. First, it situates quantum in Japan's evolving security environment. Second, it outlines the policies, institutions and businesses that make up Japan's quantum ecosystem. Third, it examines quantum as an extension of developmental state practices into the security domain. Finally, it assesses the strategic challenges and policy implications that follow.

2. Japan's Evolving National Security Context

The 2020s have been marked by intensifying Sino-U.S. rivalry over advanced technologies.[9] Washington's 2018 and 2022 *National Defense Strategies* and the 2022 CHIPS and Science Act explicitly identify semiconductors, quantum computing and artificial intelligence as areas of strategic competition.[10] China, meanwhile, has invested heavily in quantum research. In 2016 the Chinese Academy of Sciences launched the quantum satellite Micius.[11] It has rolled out national quantum communication networks, and it set ambitious targets in its 14th Five-Year Plan.[12] Japan's position concerning this rivalry is delicate. As a U.S. treaty ally and a member of the G7 it shares U.S. concerns about the technological ascent of China. However, China remains Japan's largest trading partner, which means there is deep economic interdependence. This tension has made not just economic growth but also economic security central concerns for Japanese security policymakers.

The passage of the Economic Security Promotion Act (Keizai Anzen Hoshō Suishin Hō) in 2022 institutionalised the linkage between technology and national security.[13] The law has four pillars: strengthening supply chain resilience, in particular reducing dependence on China for rare earths and semiconductors; safeguarding critical infrastruc-

7 Linda Weiss, *America Inc.? Innovation and Enterprise in the National Security State* (Ithaca, NY: Cornell University Press, 2014).

8 Alexandra Sakaki and Sebastian Maslow, "Japan's New Arms Export Policies: Strategic Aspirations and Domestic Constraints," *Australian Journal of International Affairs* 74, no. 6 (2020): 649-69, https://doi.org/10.1080/10357718.2020.1781789.

9 M. Taylor Fravel, "Technological Competition and the US-China Rivalry," in Not Just Another Cold War: The Global Implications of the US-China Rivalry, ed. Bård Nikolas Vik Steen (New York: Oxford University Press, 2025), 149-174.

10 For the 2018 National Defense Strategy see https://media.defense.gov/2020/May/18/2002302061/-1/-1/1/2018-NATIONAL-DEFENSE-STRATEGY-SUMMARY.PDF, and for the 2022 document https://media.defense.gov/2022/Oct/27/2003103845/-1/-1/1/2022-NATIONAL-DEFENSE-STRATEGY-NPR-MDR.pdf. For the full text of the CHIPS Act see https://www.congress.gov/bill/117th-congress/house-bill/4346.

11 Karen Kwon, "China Reaches New Milestone in Space-Based Quantum Communications," *Scientific American*, 15 June 2017, https://www.scientificamerican.com/article/china-reaches-new-milestone-in-space-based-quantum-communications/; Elsa B. Kania, "China's Quantum Future: Xi's Quest to Build a High-Tech Superpower," *Foreign Affairs*, 26 September 2018, https://www.foreignaffairs.com/articles/china/2018-09-26/chinas-quantum-future.

12 The State Council of the People's Republic of China, "China to Include Quantum Technology in its 14th Five-Year Plan," 22 October 2020, https://english.www.gov.cn/news/videos/202010/22/content_WS5f90e700c6d0f7257693e3fe.html.

13 Economic Security Promotion Act, https://www.japaneselawtranslation.go.jp/en/laws/view/4523/en.

ture in energy, telecommunications and transport; supporting the development of critical technologies such as quantum, AI and biotechnology; and protecting sensitive patents with dual-use potential. Quantum computing is explicitly listed among the technologies eligible for state support. The Act provides budgetary allocations, legal protection and secrecy measures to channel R&D toward national aims. In parallel, a National Security Strategy released in December 2022 by the Kishida administration elevated economic security to a central pillar of national strategy, which highlights supply chain security and in particular the vulnerability of advanced semiconductor production.[14] Together, these measures show how economic and security concerns have become intertwined. A Quantum Technology Innovation Strategy introduced in January 2020 by the Abe administration and overseen by the Integrated Innovation Strategy Promotion Council of the Cabinet Office complements this broader agenda by setting specific roadmaps for quantum computers, communication, sensing and post-quantum cryptography.[15] While the overall strategy is Cabinet Office-led, METI plays a central role in implementing its industrial policy dimension by fostering public-private collaboration and supporting commercialisation efforts.[16]

Although the MOD has not publicly announced where quantum technologies will be employed, it is clearly elevating quantum R&D as a strategic priority. An official briefing of the Cabinet Office Council for Science, Technology and Innovation (CSTI) describes quantum as a technology that could "fundamentally change future ways of war," and lays out a development line from an exploratory study to equipmentisation in the 2023 Defence Technology Guidelines.[17] Fiscal year 2023 allocations covered (i) survey research on the applicability to defence of gate-type and annealing-type quantum computers and quantum sensing, (ii) basic research funded by the Security Technology Research Promotion programme, e.g. quantum-interference-based clock stabilisation, long-distance undersea/underground quantum sensing, portable solid-state light sources for wireless quantum key distribution (QKD) and (iii) bridge research on equipmentisation, e.g. high-performance non-GNSS navigation, physical-layer secrecy for RF links and high-operability quantum magnetometers. This brief also notes continued requests in fiscal year 2024 along the same three lines.

Institutionally, in 2024 the MOD Acquisition, Technology & Logistics Agency (ATLA) instructed the Defence Innovation Science & Technology Research Institute (Bōei Inobēshon Kagaku Gijutsu Kenkyūjo) – widely described as a 'Japan-style DARPA' – to accelerate the transition from breakthrough science to capability and to run rapid challenge-driven programmes using external talent.[18] Recent budget overviews show scaling up across the portfolio: in fiscal year 2025 Security Technology Research Promotion (¥11.4bn), Breakthrough Research (¥20.1bn) and Advanced Technology Bridging Research (¥17.5bn) all aimed to employ novel functions in defence use.[19] ATLA is also using alliance-linked innovation mechanisms. For example, it co-hosted the 2025 U.S.-Japan Global Innovation Challenge with the U.S. Defense Innovation Unit to develop dual-use solutions.[20] In sum, these moves show that the quantum focus of the MoD is not a one-off. It is now part of a broader well-resourced line of development from basic research and PQC/QKD-relevant work to bridge projects and challenge programmes, which are consistent with Japan's securitised developmental

14 Cabinet Office, National Security Strategy, https://www.cas.go.jp/jp/siryou/221216anzenhoshou/nss-e.pdf.

15 Cabinet Office, Quantum Technology Innovation Strategy, https://www8.cao.go.jp/cstp/tougosenryaku/ryoushisenryaku_r.pdf.

16 For example, METI, Ryōshi konpyūta no sangyōka ni muketa kaihatsu no kasoku oyobi kankyō seibi jigyō (Project for Accelerating Development and Establishing an Environment for the Industrialisation of Quantum Computers), 2025, https://www.meti.go.jp/information_2/publicoffer/review2025/kokai/0602_3gaiyo.pdf.

17 MOD, Ryōshi kanren no torikumi, 21 September 2023, https://www8.cao.go.jp/cstp/ryoshigijutsu/16kai/siryo3-6.pdf.

18 ATLA, November 2024, https://www.mod.go.jp/atla/research/ats2024/pdf_oral_matl/1112_1115_ss03.pdf?utm_source=chatgpt.com; Tokyo Shimbun, 13 April 2024, https://www.tokyo-np.co.jp/article/320947.

19 MOD, Progress and Budget in Fundamental Reinforcement of Defence Capabilities: Overview of FY2025 Budget, https://www.mod.go.jp/en/d_act/d_budget/pdf/fy2025_20250411a.pdf?utm_source=chatgpt.com.

20 ATLA, n.d., https://www.mod.go.jp/atla/disti_inno_chal_2025.html.

state approach to steering next-generation technologies.

The most immediate security implication of quantum computing lies in the realm of cybersecurity. A sufficiently advanced quantum computer can make widely used encryption systems, such as RSA and elliptic curve cryptography, obsolete. This so-called 'quantum decryption' threat has already prompted adversaries to engage in 'harvest now, decrypt later' strategies, storing encrypted communications today in anticipation of future breakthroughs. Japan has recognised this risk and it incorporated quantum in its 2021 Cybersecurity Strategy.[21] This strategy has two parallel objectives: to develop post-quantum cryptography (PQC), with agencies such as the Information-Technology Promotion Agency (IPA) and the National Institute of Advanced Industrial Science and Technology (AIST) testing algorithms in line with U.S. NIST standards; and to deploy QKD networks, in which NICT has been a global pioneer since 2010, and concerning which firms such as Toshiba, NEC and NTT are now commercialising secure systems for critical infrastructure and financial services.[22]

These efforts show how economic and security logics converge. PQC strengthens the resilience of national infrastructure while QKD offers Japan a potential commercial advantage in secure communications – an industrial policy consistent with the country's developmental state tradition. However, tensions remain. For instance, over-securitising cryptographic research could restrict academic openness and stifle innovation, while insufficient securitisation could expose Japan to strategic vulnerabilities. Therefore, the challenge is to strike a balance, i.e. to contribute to open international standards while protecting sensitive applications in defence, finance and infrastructure.[23] Cybersecurity is therefore the reason for the present-day urgency of Japan's quantum policy. Unlike speculative breakthroughs in hardware that may not materialise for decades, the cryptographic implications of quantum computing demand action today. Having outlined the policy and security context in which Japan situates its quantum ambitions, the next section examines the ecosystem in which these ambitions are pursued in practice.

3. Japan's Quantum Ecosystem: Policies, Institutions and Industry

Japan's quantum computing landscape is sustained by a dense web of public research institutions, major corporations and emerging startups. This ecosystem reflects both the enduring strengths of Japan's research base and the challenges involved in mobilising it for national security objectives.

At the centre of policy coordination is the Cabinet Office Quantum Technology Innovation Council (Ryōshi Gijutsu Inobēshon Kaigi), which was established in 2019 under the Abe administration. Within METI, the Innovation Policy Division Quantum Industry Office (Inobēshon Seisaku-ka Ryōshi Sangyō-shitsu) oversees industrial policy in this field. At MEXT, responsibilities are split between the Quantum Research Promotion Office (Ryōshi Kenkyū Suishin-shitsu) and the Computational Science and Technology Promotion Office (Keisan Kagaku Gijutsu Suishin-shitsu). Recently, the Ministry of Defence has expanded its role by establishing the Defence Innovation Science and Technology Institute (Bōei Inobēshon Gijutsu Kenkyūjo, DISTLI) in the ATLA, given the growing securitisation of quantum research.

Public research institutions form the backbone of Japan's quantum R&D. RIKEN hosts the Centre for Quantum Computing (RQC)[24] in Wako, a flagship in superconducting qubit research in partnership with IBM, and it plays a leading role in international consortia such as the Quantum Innovation Initiative Council (QII).[25] The National Institute of Information and Communications Technology pioneered QKD and secure satellite communica-

21 NISC, Japan's Cybersecurity Strategy 2021, 28 September 2021, https://www.nisc.go.jp/pdf/policy/kihon-s/cs-senryaku2021-gaiyou-en.pdf.

22 *Nihon Keizai Shimbun*, 28 July 2025, https://www.nikkei.com/article/DGXZQOSG25AZ50V20C25A7000000/.

23 See also Dai Mochinaga, "Japan's Shift in the Cyber Domain toward a Proactive Defense Posture," *Asia Policy* 20, no. 2 (2025): 35-45.

24 https://www.riken.jp/en/research/labs/rqc/.

25 https://qii.jp. *Nikkei Asia*, "IBM and Japan institute team up to develop next-gen quantum computer," 16 June 2024, https://asia.nikkei.com/business/technology/ibm-and-japan-institute-team-up-to-develop-next-gen-quantum-computer.

tions and laid the foundation for a future quantum internet. Universities also play critical roles. The University of Tokyo is prominent in superconducting qubits and quantum chemistry, Osaka University in ion traps and quantum optics, Kyoto University in quantum information theory and Keio University in algorithms. Partnering with Intel, the National Institute of Advanced Industrial Science and Technology (AIST) connects quantum with applied materials science and metrology.[26] These institutions serve not only as research hubs but also as bridges between the government, corporations and startups, thus echoing the structure of earlier Japanese technology consortia.

Major corporations remain central players. NEC, a pioneer in superconducting qubits since the late 1990s, is now developing hybrid quantum-classical systems.[27] Fujitsu has invested in quantum computing through its digital annealers and collaborates with RIKEN on quantum simulators.[28] NTT focuses on optical quantum computing and communication, and contributes to large-scale QKD demonstrations. Toshiba has emerged as a global leader in QKD commercialisation. It exports secure communication technologies to markets in Europe and Asia.[29] These firms benefit from long-term subsidies, preferential procurement and integration in national strategies – classic Japanese industrial policy instruments.

Development of quantum hardware is inseparable from the semiconductor ecosystem. Most platforms – superconducting qubits, trapped ions integrated with silicon chips and photonic circuits – depend on advanced semiconductor manufacturing. Japan's position here is mixed. It has lost ground in design and leading-edge manufacturing to Taiwan and South Korea but retains dominance in upstream inputs such as silicon wafers (Shin-Etsu Chemical, SUMCO), photoresists (JSR, Tokyo Ohka Kogyo), and specialty gases. Tokyo Electron remains a leading supplier of semiconductor manufacturing equipment. These strengths are vital for both conventional and quantum manufacturing. However, dependence on Taiwan for cutting-edge logic chips exposes Japan to strategic vulnerabilities. In response, Tokyo has moved aggressively to rebuild domestic capacity.[30] Its 2024 semiconductor revitalisation strategy[31] is a comprehensive agenda to restore competitiveness and strengthen supply chain resilience, and it complements major subsidies of a TSMC plant in Kumamoto by Sony and Denso.[32] At the same time, the government has backed Rapidus – a state-supported consortium including Toyota, Sony, NTT and others tasked with producing next-generation two-nanometre chips in Hokkaidō by the late 2020s.[33] Together, these initiatives reflect how supply chain resilience has been securitised. For quantum, semiconductor stability ensures manufacturing capacity, shields against bottlenecks and aligns Japan with allied diversification drives such as the U.S. CHIPS and Science Act and the EU Chips Act.

Startups, often spun out of universities, bring agility to the ecosystem. QunaSys develops algorithms for chemical and material simulations and partners with global firms. Blueqat provides software platforms to broaden access to quantum computing through the cloud. Sigma-i, a spin-off from the Tokyo Institute of

26 *Nikkei Asia*, "Japan lab and Intel set out to develop next-gen quantum computer," 6 February 2025, https://asia.nikkei.com/business/technology/japan-lab-and-intel-set-out-to-develop-next-gen-quantum-computer.

27 https://www.nec.com/en/global/rd/technologies/202307/index.html.

28 *Nikkei Asia*, "Fujitsu's new quantum computer to vie for world's most powerful," 1 August 2025, https://asia.nikkei.com/business/technology/fujitsu-s-new-quantum-computer-to-vie-for-world-s-most-powerful.

29 See https://www.soumu.go.jp/main_content/000976596.pdf.

30 Kazuto Suzuki, "In chip renaissance, Japan is learning from past mistakes," Institute of Geoeconomics, 5 March 2025, https://instituteofgeoeconomics.org/en/research/in-chip-renaissance-japan-is-learning-from-past-mistakes/; The Economist, "Japan storms back into the chip wars," 21 August 2025, https://www.economist.com/asia/2025/08/21/japan-storms-back-into-the-chip-wars.

31 METI, Outline of Semiconductor Revitalization Strategy in Japan, July 2024, https://www.meti.go.jp/english/policy/0704_001.pdf.

32 *The Economist*, "TSMC could revolutionise rural Japan," 21 August 2025, https://www.economist.com/asia/2025/08/21/tsmc-could-revolutionise-rural-japan.

33 *The Japan Times*, "Rapidus reports 2-nanometer progress, moving Japan closer to semiconductor goals," 18 July 2025, https://www.japantimes.co.jp/business/2025/07/18/companies/rapidus-advanced-chip/.

Technology, focuses on optimisation problems using quantum annealing. Recognising the importance of these firms, the Japanese government has committed approximately 50 billion yen (about $335 million USD) to accelerate the industrialisation of its domestic quantum technology sector, with funding directed to scaling startups and fostering commercialisation. This initiative reflects a broader attempt in 2025 by Ishiba and METI to position Japan as a global leader in quantum computing, tying entrepreneurial activity to national strategy.[34]

However, observers have noted limitations of this approach. Analysts caution that, while government support is symbolically important, the resources allocated remain modest when compared to the multi-billion-dollar programmes in the United States, China and the EU. They warn that Japan risks spreading funding too thinly across multiple initiatives without providing the sustained large-scale backing needed to keep pace with its global competitors.[35] This critique highlights a familiar tension in Japan's techno-industrial policy: the state can effectively mobilise resources and coordinate actors but its financial commitments may not always match its strategic ambitions. Even in the entrepreneurial domain, startups remain heavily reliant on state support and corporate partnerships, revealing both the persistence and the limits of developmental state practices.

Finally, international collaboration anchors Japan's quantum strategy in its alliance and partnership networks. A 2021 U.S.-Japan Competitiveness and Resilience (CoRe) Partnership[36] launched joint projects in quantum, AI and semiconductors, and embedded Tokyo in the U.S. technological ecosystem. A 2023 G7 Hiroshima Leaders Communiqué[37] reaffirmed collective commitments to open standards and responsible use. Japan and the EU have also expanded their ties with economic and strategic partnership agreements, with quantum included in science and technology dialogues. Recent initiatives are part of this momentum. The European Parliament STOA Panel has hosted the 8th STOA-STS Forum Conference on EU-Japan cooperation on artificial intelligence and quantum technologies, reflecting the growing political salience of these fields.[38] Through Horizon Europe, Japanese institutions have joined EU quantum consortia, in particular the Quantum Flagship.[39] Toshiba Europe and its Japanese partners have collaborated with European institutes on QKD pilots in finance, while both sides coordinate in ISO and ITU bodies to foster open standards. Academic exchanges supported by Horizon Europe and the Japan Society for the Promotion of Science (JSPS) further strengthen talent mobility.

Japan-EU cooperation reflects a policy logic different to that of the U.S.-Japan partnership. Whereas ties with Washington are more closely linked to defence and alliance politics, EU-Japan projects emphasise open frameworks, privacy protection and balancing security with innovation. However, the integration of Japan with the U.S. ecosystem remains far deeper, with EU ties being fragmented and underfunded. Regulatory differences, such as that between the GDPR and Japan's more flexible data regime, complicate collaboration. In sum, Japan's quantum ecosystem is embedded in global alliances but constrained by geopolitical rivalry – a sharp contrast to the more open industrial networks of the 1980s.

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34 Matt Swayne, "Japan Backs Quantum Startups In 50 Billion Yen Push For Tech Leadership," *Quantum Insider*, 31 July 2025, https://thequantuminsider.com/2025/07/31/japan-backs-quantum-startups-in-50-billion-push-for-tech-leadership/.

35 Wakana Asano, "Japan needs to take the quantum-technology leap," *The Japan Times*, 9 July 2025, https://www.japantimes.co.jp/commentary/2025/07/09/japan/japan-quantum-technology-leap/.

36 MOFA, U.S.-Japan Competitiveness and Resilience (CoRe) Partnership, https://www.mofa.go.jp/files/100177722.pdf.

37 MOFA, 20 May 2023, https://www.mofa.go.jp/policy/economy/summit/hiroshima23/documents/pdf/Leaders_Communique_01_en.pdf.

38 European Parliament, "EU-Japan: Shaping the future of AI and quantum technologies," 11 April 2025, https://epthinktank.eu/2025/04/11/eu-japan-shaping-the-future-of-ai-and-quantum-technologies/; European Commission, "EU and Japan strengthen Research and Innovation cooperation in quantum science and technology," 13 May 2025, https://digital-strategy.ec.europa.eu/en/news/eu-and-japan-strengthen-research-and-innovation-cooperation-quantum-science-and-technology.

39 Cabinet Office, Signing of the Letter of Intent on Japan-EU Cooperation in Quantum Science and Technology, 13 May 2025, https://www8.cao.go.jp/cstp/stmain/20250513ryoshi_en.html.

4. Quantum as Industrial Policy and the Developmental State 2.0

Quantum computing in Japan is best understood through the lens of the developmental state – a model in which the state orchestrates industrial transformation through targeted policies, bureaucratic guidance and close public-private cooperation.

Japan's current quantum policy bears a striking resemblance to earlier episodes of state-led technological development. In the 1970s MITI organised the VLSI Project, which pooled resources from Fujitsu, NEC, Hitachi and Toshiba to close the gap with U.S. firms. This consortium propelled Japan to global semiconductor leadership in the 1980s. In the 1980s and 1990s the government backed supercomputing projects such as the Earth Simulator, which symbolised national prestige and technological ambition. In the 2000s targeted support was directed toward robotics and nanotechnology, sectors justified as drivers of economic growth. With these initiatives the state acted as a strategic investor and coordinator of private firms, embodying what scholars such as Marie Anchordoguy have termed "communitarian capitalism,"[40] a system in which collaboration and long-term relationships are privileged over market liberalism.

Today, Japan's developmental state has adapted to new realities. Whereas earlier industrial policy emphasised competitiveness and economic growth, contemporary strategies are explicitly framed as economic security measures designed to protect national survival.[41] The quantum agenda is also closely intertwined with the U.S. alliance. Unlike the relative autonomy of the 1970s, Japan now operates within an alliance framework and embeds its policies in the broader context of Sino-U.S. rivalry. Furthermore, Japan's quantum ecosystem depends on global research networks, mobility of talent and norms of open science, which creates tensions with securitisation. The result is a 'developmental state 2.0,' which is characterised by securitised techno-nationalism, alliance-driven R&D and persistent state coordination.

In pursuing quantum leadership, Japan is deploying familiar industrial policy instruments in a new context. METI continues to allocate subsidies and R&D grants to startups and consortia, while the Cabinet Office oversees cross-ministerial strategies in a role reminiscent of the earlier coordination by MITI.[42] Public procurement remains an important mechanism, with government agencies acting as early markets for quantum technologies such as secure communications. Japan also actively participates in international standard-setting bodies, including ISO and ITU, thus extending industrial policy into global governance by helping to shape the norms of quantum technologies. This reflects continuity, but it is also adapted to an environment in which security imperatives weigh as heavily as economic considerations.

Theoretically, Japan's quantum strategy challenges the notion that the developmental state has become obsolete. Rather, it demonstrates the persistence of state-led coordination, albeit with a new rationale. The justification for intervention has shifted from promoting industrial growth and catching up with the West to ensuring national security and sustaining relevance in an era of techno-strategic competition. Quantum computing is an example of this fusion of economic and security imperatives, which produces a new form of techno-industrial governance. Japan's quantum strategy embodies the emergence of a securitised developmental state in which technological policy is simultaneously industrial, scientific, and strategic.

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40 Marie Anchordoguy, *Reprogramming Japan: The High Tech Crisis under Communitarian Capitalism* (Ithaca: Cornell University Press, 2005).

41 Ali Hayes and Trissia Wijaya, "State Transformation and Social Forces Under the Shadow of Economic Security: The Case of Japan," *The Asia-Pacific Journal/Japan Focus*, 18 December 2023, https://apjjf.org/2023/21/wijaya-hayes.

42 See also Greogory W. Noble, *Japan's New Industrial Policy* (Cambridge: Cambridge University Press, 2025).

5. Strategic Challenges and Risks

While on paper Japan's quantum ecosystem appears robust, implementation of its strategy faces challenges that could undermine its ambitions. These risks reflect both structural constraints and policy dilemmas that shape the trajectory of Japan's quantum future.

Japan still lags behind the U.S. and China. The United States benefits from deep venture capital markets, world-leading university research clusters and companies such as IBM, Google and Microsoft at the forefront of quantum hardware development. China, meanwhile, has mobilised billions of dollars in state-led investment and showcased world-first achievements in satellite-based quantum communication.[43] Japan's progress – although tangible – remains incremental. For instance, NEC and RIKEN have demonstrated superconducting qubit prototypes,[44] but none of them yet approach the scale of Google's 53-qubit Sycamore or IBM's 127-qubit Eagle. This gap raises questions about whether Japan can sustain global competitiveness without aligning more deeply with U.S. platforms.

Like the United States, Japan faces a persistent shortage of skilled personnel.[45] Expanding STEM enrolments at universities and securing stable career pathways for young researchers are critical priorities. It is equally important to remove barriers that hinder the recruitment of foreign talent, such as stringent language requirements, bureaucratic procedures and restrictive immigration policies. Without a more robust human capital strategy, Japan risks deepening its dependence on foreign research ecosystems.[46] In recognition of this challenge, Tokyo and Washington have launched a U.S.-Japan University Partnership for Workforce Advancement and Research & Development in Semiconductors (UPWARDS) for The Future, a consortium linking six leading Japanese universities with five U.S. counterparts to foster joint research, student exchanges and workforce training in semiconductor and quantum-related fields.[47] These initiatives show that talent development is not only a national priority but also an area in which alliance cooperation can mitigate domestic shortfalls.

Japan also faces a dilemma of balancing openness and security. Quantum innovation depends on open science, international collaboration and shared data. However, Japan's economic security agenda emphasises secrecy, investment screening and export controls.[48] This tension is particularly acute in academia, in which researchers fear restrictions on international cooperation. If policies lean too heavily toward securitisation, innovation could be stifled and foreign researchers discouraged from working with Japanese institutions. Conversely, if openness prevails there is a risk of sensitive intellectual property leaking to strategic competitors. Managing this trade-off is one of the most difficult dilemmas in Japan's techno-security strategy.

A central danger is a drift toward techno-nationalism.[49] A securitised developmental state could subordinate technology policy entirely to national security, thus undermining Japan's comparative

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43 Antonia Hmaidi and Jeroen Groenewegen-Lau, "China's long view on quantum tech has the US and EU playing catch-up," Merics, 12 December 2024, https://merics.org/en/report/chinas-long-view-quantum-tech-has-us-and-eu-playing-catch.

44 *Nikkei Asia*, "Fujitsu's new quantum computer to vie for world's most powerful," 1 August 2025, https://asia.nikkei.com/business/technology/fujitsu-s-new-quantum-computer-to-vie-for-world-s-most-powerful.

45 Sam Howell, "The United States' Quantum Talent Shortage Is a National Security Vulnerability," *Foreign Policy*, 31 July 2023, https://foreignpolicy.com/2023/07/31/us-quantum-technology-china-competition-security/; Cabinet Office, Ryōshi sangyō no sōshutsu/hatten ni muketa suishin hōsaku (Promotion measures for the creation and development of the quantum industry), 9 April 2024, https://www8.cao.go.jp/cstp/ryoshigijutsu/240409_q_measures.pdf; Cabinet Office, Ryōshi ekoshisutemu kōchiku ni muketa suishin hōsaku gaiyō (Outline of promotion measures for building a quantum ecosystem), 30 May 2025, https://www8.cao.go.jp/cstp/ryoshigijutsu/250530_q_ecosys_gaiyo.pdf.

46 Rob Stroeks, "Quantum leaps in Japan create opportunities for the Netherlands," Netherlands Enterprise Agency, July 2025, https://www.rvo.nl/sites/default/files/2025-07/JapanQuantumLeaps.pdf.

47 Brad Glosserman, "Semiconductors are back to center stage in the Japan-U.S. alliance," *The Japan Times*, 5 March 2024, https://www.japantimes.co.jp/commentary/2024/03/05/japan/semiconductors-japan-us-alliance/.

48 Sae Kobayashi, "Japan's Export Control on Quantum Technology," *Stanford International Policy Review*, 12 September 2024, https://fsi.stanford.edu/sipr/japan-qt.

49 Bronte Munro, "The U.S. Can't Lead on Quantum Computing Alone," *Foreign Policy*, 25 October 2023, https://foreignpolicy.com/2023/10/25/quantum-computing-united-states-australia-cooperation-allies-science-technology-chips/.

advantage as an open innovation-friendly democracy. Over-broad export controls might curtail access by Japanese firms to global markets, while a full severing of collaboration with China could isolate Japan's research community.[50] The challenge is to advance economic security without repeating the insular mistakes of the 1990s, when protectionist responses to globalisation weakened Japan's high-tech industries.

Recent U.S. policies illustrate these risks. The drive by the Trump administration to reshore semiconductor manufacturing – combined with tariffs on chip imports and unilateral export restrictions – disrupts global supply chains and complicates international research and development. Therefore, for Japan excessive reliance on a U.S.-driven securitised approach could similarly constrain innovation, limit international partnerships and expose domestic firms to geopolitical shocks.

Japan's dependence on the United States therefore creates vulnerabilities of its own. The alliance secures access to cutting-edge research and ensures defence interoperability, but over-reliance risks eroding strategic autonomy. Already, many flagship quantum projects are tied to U.S. firms, such as the RIKEN-IBM partnership and Toshiba's QKD exports to American allies. Should U.S. policy shift – as it has done under the Trump administration – Japan could face renewed supply shocks or be pressured into restrictive measures that prioritise American interests. Safeguarding relative independence within the alliance is therefore a strategic imperative.

6. Policy Implications and Recommendations

Japan must refine its quantum strategy in ways that balance industrial development, national security and global cooperation. Domestically, this requires strengthening the supply of talent with expanded STEM curricula, fellowships and stable career paths, while easing immigration rules to attract highly skilled scientists. Supporting startups is equally vital, with public-private venture funds and government procurement providing stable markets. Better coordination is also needed. A single 'Quantum Technology Agency' could streamline overlapping programmes in METI, MEXT and the Cabinet Office, while stronger university-industry links would accelerate technology transfer.

At the same time, Japan must strike a balance between secrecy and openness, and protect sensitive defence research with tiered classifications without over-regulating basic science, and foster 'secure openness' by embedding researchers in controlled but internationalised networks.

At the alliance level, Japan should deepen cooperation with the United States and the EU with joint research centres and a dedicated quantum security forum on cryptography, QKD and standards. Export control policies must be harmonised with Washington and Brussels, but in a manner that preserves flexibility for Japanese firms and prevents unilateral U.S. and EU measures from over-constraining domestic research.

Finally, Japan must guard against over-dependence on U.S. partners by investing in independent capabilities to ensure resilience should American support falter. Globally, Japan has an opportunity to promote multilateral norms through the G7, OECD and CPTPP by encouraging open standards and responsible dual-use management. It should also strengthen partnerships with other democratic technology leaders – including the EU, South Korea, Canada and Australia – on R&D, talent exchange and export control regimes.

Japan is well placed to convene both advanced and developing economies, and so to help prevent fragmentation of the quantum ecosystem into rival blocs and reduce the risk of a quantum arms race.

7. Conclusion

Quantum computing stands at the intersection between technological innovation and geopolitical competition. For Japan, it offers both the possibility of reclaiming leadership in advanced technology and the challenge of adapting its developmental state to new strategic realities. The country's current approach reflects the logic of a securitised developmental state. It deploys industrial policy tools, mobilises universities and corporations, and ties economic vitality to national security imperatives. This is consistent with Japan's postwar develop-

50 Kazuyuki Motohashi, "U.S.-China High Tech Friction and Japan's Response: Implications from comparison of innovation processes in Japan, the United States, and China," REITI Policy Update 121, 7 February 2025, https://www.rieti.go.jp/en/special/policy-update/121.html.

mental trajectory yet adapted to the pressures of U.S.-China rivalry and globalised innovation.

Success, however, is not assured. Japan must address structural disadvantages such as lagging hardware capabilities, limited human capital and heavy alliance dependence while carefully balancing openness with security. A drift toward techno-nationalism could isolate its researchers and firms and so undermine its competitiveness.

If Japan can manage these trade-offs, quantum computing may emerge not only as a pillar of national security resilience but also as a symbol of renewed technological relevance. The broader lesson is clear: industrial policy is not obsolete. In the quantum era, it is more central than ever to both national security and economic vitality.

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